MIL-V-62358(AT)
11 May 1983
Superseding
(see 6.7)

MILITARY SPECIFICATION

VALVE, SOLENOID

This specification is approved for use by the US Army Tank-Automotive Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers an electrically operated solenoid valve used in the transmission to select gear range by porting oil to required control valves.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Tank-Automotive Command, ATTN: DRSTA-GSS, Warren, MI 48090, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter

FSC 2520

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SPECIFICATIONS
MILITARY

MIL-L-2104	- Lubricating Oil, Internal Combustion Engine, Tactical Service.
MIL-L-21260	- Lubricating Oil, Internal Combustion Engine, Preservative and Break-in.
MIL-L-46167	- Lubricating Oil, Internal Combustions Engine, Arctic.

STANDARDS
MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-810	- Environmental Test Methods.

2.1.2 Other Government documents, Government drawings, and publications. The following documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS
ARMY

12267989	- Oil Filter Element.
12288178	- Solenoid Assembly, 24 Volt.

(Copies of specifications, standards, and drawings required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity, or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Preproduction. When specified (see 6.2), the contractor shall furnish sample units for preproduction inspection and approval (see 4.4.1 and 6.3). Preproduction samples shall be inspected by the contractor under the surveillance of the Government to determine conformance to the requirements of this specification. Preproduction samples shall be fully representative of units to be supplied from production tooling and facilities.

3.2 Initial production. Unless otherwise specified (see 6.2), initial production units shall be furnished. Units shall be examined and tested to determine conformance to all requirements of this specification. If submitted units meet all requirements of this specification, no modifications

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shall be applied to subsequent units to be produced under the contract without prior approval of the Government.

3.3 Materials. Materials shall be as specified herein and in referenced specifications, standards and drawings. Material shall be free of defects which adversely affect performance or serviceability of the finished product (see 6.6).

3.4 Construction. Units shall conform to Drawing 12288178 and applicable component drawings and Military standards.

3.5 Performance.

3.5.1 Coil resistance. Resistance of solenoid valve coil at 70^0F $\pm$ 5^0F shall be 60 to 70 ohms measured between solenoid connector and ground (see 4.6.1).

3.5.2 Pull-in-voltage. Solenoid valve shall open at not more than 18 volts and reclose when voltage is reduced to not less than 1 volt (see 4.6.2).

3.5.3 Bleed rate. Pressure in the cavity between the feed orifice and solenoid valve shall not exceed 18 psi (see 4.6.3).

3.5.4 Valve leakage. Valve leakage shall not exceed a 10 psi drop (see 4.6.4).

3.5.5 Planar function. Assembly shall meet the requirements of 3.5.3 when mounted in any of the following planes (see 4.6.5).

a. Flange down.
b. Flange up.
c. Bottom of the flange vertical, mounting holes horizontal and the lead coming out the top.

3.5.6 Physical characteristics.

3.5.6.1 Endurance. Assembly shall meet, without failure, the requirements of 3.5.1 through 3.5.5 during and after subjection to 500,000 equal time, on and off cycles, at a rate of 60 $\pm$ 5 cycles per minute (see 4.6.6.1).

3.5.7 Environmental conditions.

3.5.7.1 Low temperature. After submersion in oil conforming to MIL-L-46167 at -60 $\pm$ 5^0F for 24 $\pm$ 1 hours, valve shall open at not more than 18 volts at which time pressure in a cavity between orifice and valve shall not exceed 30 psi. Valve shall close when voltage is reduced to not less than 1 volt at which time pressure drop across orifice shall be not more than 10 psi. Assembly shall meet requirements of 3.5.1 through 3.5.5 following which assembly lead wire shall withstand a 180 $\pm$ 5^0 bend around a 0.75 $\pm$ .02 inch diameter arbor without cracking (see 4.6.7.1).

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3.5.7.2 High temperature. After submersion in oil conforming to MIL-L-21260, Grade 30 which conforms to MIL-L-2104, Grade 30 oil with 135 to 270 ppm biocide, at 300 ± 10^0F for 24 hours, valve shall open at not more than 18 volts at which time pressure in a cavity between orifice and valve shall not exceed 18 psi. Valve shall close when voltage is reduced to not less than 1 volt at which time pressure in a cavity between orifice and valve shall be not less than 80 psi. Assembly shall meet the requirements of 3.5.1 through 3.5.5, following which assembly lead wire shall withstand a 180 ± 50 bend around a 0.75 ± .02 inch diameter arbor without cracking (see 4.6.7.2).

3.5.7.3 Thermal shock. Assembly shall withstand 15 thermal shock cycles of a non-energized soak of not less than 2 hours at -60 ± 5^0F and a soak of not less than 1 hour at 300 ± 5^0 F while energized with +30 ± 1 volt, following which assembly shall meet requirements of 3.5.1 through 3.5.5 (see 4.6.7.3).

3.5.7.4 Basic shock. Assembly shall meet requirements of 3.5.1 through 3.5.5 after exposure to shock impulses of 30 ± 3g, at 11 ± 1.1 milliseconds (ms) half sine wave applied in each direction of 3 mutually perpendicular axes (see 4.6.7.4).

3.5.7.5 Gun firing shock. Assembly shall meet requirements of 3.5.1 through 3.5.5 after exposure to gun firing shock of 55 ± 5.0g for 2.5 ± 0.2 ms duration applied along equivalent transmission transverse and longitudinal axis (see 4.6.7.5).

3.5.7.6 Sinusoidal vibration. Assembly shall meet the requirements of 3.5.1 through 3.5.5 after exposure to sinusoidal vibration in accordance with figure 1 in each of three mutually perpendicular axes. Vibration shall be imposed at a logarithmic sweep rate of 12 ± 1 minutes per sweep cycle from 7 to 2700 to 7 hertz (Hz), for 28 sweep cycles in each of three mutually perpendicular axes. An equal number of sweeps will be made, both energized to +30 ± 1 volts and de-energized, along each axis (see 4.6.7.6.1).

3.5.7.7 Random vibration. Assembly shall meet the requirements of 3.5.1 through 3.5.5 after being subjected to 100 hours of random vibration durability testing in any one of three mutually perpendicular axes. Test conditions shall be as specified in the random vibration spectrum curve in figure 2. Assembly shall be energized with +30 ± 1 volts and cycled at a rate of 15 minutes on and 15 minutes off. Test may be performed at room ambient conditions (see 4.6.7.6.2).

3.6 Identification and marking. Identification and marking shall be in accordance with MIL-STD-130 and as noted on Drawing 12288178.

3.7 Workmanship. Workmanship shall meet all specified requirements and shall assure that solenoid valve assemblies are free of defects that would affect performance.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection equipment. Unless otherwise specified in the contract (see 6.2), the supplier is responsible for the provision and maintenance of all inspection and test equipment necessary to assure that supplies and services conform to contract requirements. Commercial, modified commercial, or supplier designated inspection equipment or measuring set-ups must be capable of repetitive measurements to an accuracy of 10 percent of the component tolerance.

4.1.2 Materials. The contractor's inspection records shall be examined to determine contractor conformance to 3.3.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Preproduction inspection (see 4.4.1).
b. Initial production inspection (see 4.4.2).
c. Quality conformance inspection (see 4.5).

4.3 Test conditions and equipment.

4.3.1 Test conditions. Unless otherwise specified (see 6.2), all tests shall be conducted under the following conditions.

4.3.1.1 Environmental.

a.	Air temperature	40^{0} - 100^{0}F
b.	Barometric pressure	Ambient
c.	Relative humidity	Ambient

4.3.1.2 Electrical.

a. Applied voltage shall be +18 to +30 volts direct current (vdc) applied to connector.
b. Solenoid valve assembly shall be mounted on a fixture with a common negative (-) ground with the power supply.

4.3.1.3 Oil. Oil conforming to MIL-L-2104, Grade 30 shall be used in ambient temperatures between $+10^{0}$F and $+120^{0}$F. Oil conforming to MIL-L-46167 shall be used in ambient temperatures between -60^{0}F and $+20^{0}$F. Oil conforming to MIL-L-21260 which conforms to MIL-L-2104, Grade 30 with 135-270 parts per million (ppm) by weight of biocide may be used as

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an alternate to MIL-L-2104 for transmission acceptance testing, production surveillance, and break-in operation. Oil conforming to MIL-L-2104, Grade 10, or MIL-L-21260, which conforms to MIL-L-2104, Grade 10, with 135 to 270 ppm biocide shall be used for solenoid valve assembly bench testing unless otherwise specified (see 6.2).

4.3.1.3.1 Filtration. Supply oil shall be filtered to requirements of Drawing 12267989.

4.3.1.3.2 Supply pressure. Supply oil pressure shall be 180-370 psi in the transmission installation with pressures as specified in 3.5.2, 3.5.3, 3.5.4, 3.5.5 and 3.5.6.1 in solenoid valve assembly bench testing.

4.3.1.3.3 Supply temperature. Supply oil temperature shall be 175 to 265^0F in the transmission installation with temperatures as specified in 3.5.2, 3.5.3, 3.5.4, 3.5.5 and 3.5.6.1 in solenoid valve assembly bench testing.

4.3.1.3.4 Supply flow capacity. Supply oil flow capacity shall be 2.0 ± 0.1 gallons per minute.

4.3.1.4 Test equipment. Assembly test stand shall be used in the performance of the inspection and tests specified herein. Test stand shall include provisions for mounting solenoid valve assembly using installed interface configuration for its intended application (i.e. hard mounted on a one to two inch thick steel plate or equivalent). Provisions shall be made for supplying heated, filtered oil (up to 2 GPM) to solenoid valve inlet through a 0.037 ± 0.001 inch diameter by 0.25 ± 0.02 inch long feed orifice. In addition, provisions shall be made to apply electrical power across the solenoid lead wire and mounting face (positive to lead wire). Test stand shall incorporate meters and gages necessary to measure and record the following parameters during solenoid valve assembly test:

(a)	Oil supply pressure - psi	0-200 psi ± 2 percent
(b)	Oil cavity pressure - psi (between feed orifice and solenoid valve)	0-200 psi ± 2 percent
(c)	Oil temperature - ^{0}F	50^0 -275^0F ± 2 percent
(d)	Applied voltage - volts dc	0-40 vdc ± 2 percent
(e)	Solenoid coil resistance - ohms	0-100 ohms ± 2 percent

4.4 Preproduction and initial production.

4.4.1 Preproduction inspection. Twenty preproduction samples (see 3.1) shall be inspected, at a location approved by the Government, to determine conformance to requirements of this specification. Inspection shall consist of examination as specified in 4.5.2 and all tests specified in table II.

4.4.1.1 Preproduction inspection failure. Failure to pass any examination or test shall be cause for refusal to grant preproduction approval until corrective action taken by the contractor has been approved by the Government.

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4.4.2 Initial production inspection. On beginning production, twenty assemblies shall be selected from the first hundred assemblies produced, and shall undergo, and shall pass, initial production inspection (see 3.2).

4.4.2.1 Initial production test. To determine compliance with Section 3 (inclusive), assemblies will be examined as specified in 4.5.2 and tested as specified in table II. Inspections will be performed by the Government and will require no more than 90 days.

4.4.2.2 Failure. Failure of assemblies to conform to requirements during, or after, initial production test, shall be cause for rejection of assemblies. The Government may refuse to accept further production assemblies until evidence is provided by the contractor that corrective action has been taken. Defects found during, or after, tests shall be evidence that assemblies previously accepted are similarly defective, unless information to the contrary is provided by the contactor, and approved by the Government. All defects shall be corrected by the contractor at no cost to the Government.

4.5 Quality conformance inspection.

4.5.1 Sampling.

4.5.1.1 Lot formation. A lot shall consist of all assemblies from an identifiable production period, from one manufacturer, submitted at one time for acceptance.

4.5.1.2 Sampling for examination. Samples for quality conformance shall be selected in accordance with MIL-STD-105. Before sampling may be initiated, the contractor shall 100 percent inspect a minimum of 20 assemblies to establish a process average (see 6.4), to enable normal sampling in accordance with MIL-STD-105.

4.5.1.3 Sampling for acceptance testing. Each assembly from each lot (see 4.5.1.1) shall undergo acceptance tests.

4.5.2 Examinations. Assemblies shall be examined for conformance to applicable drawings and this specification. Examinations shall be performed against the classification of defects and with the acceptable quality levels (AQLs) specified in 4.5.2.1 and table I.

4.5.2.1 Acceptable quality level. Each sample selected in accordance with 4.5.1.2 shall be examined for conformance to the following acceptable quality levels (AQLS), on the basis of percent defective:

Classification	AQL
Major	1.0
Minor	2.5

4.5.2.2 Classification of defects. For examination purposes, defects shall be classified as specified in table I.

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TABLE I. Classification of defects.

Category	Defect	Method of Inspection
Major		
101	Dimensions affecting interchangeability, not within tolerance (see 3.4)	SIE 1/
102	Improper marking (see 3.6)	Visual
103	Improper workmanship, affecting inter-changeability (see 3.7)	Visual
Minor		
201	Dimensions not affecting interchangeability (see 3.4)	SIE
202	Improper workmanship affecting appearance (see 3.7)	Visual

1/ SIE = Standard Inspection Equipment

4.5.3 Classification of tests. Classification of tests shall be as follows:

a. Acceptance tests (see 4.5.4).
b. Control tests (see 4.5.5).

4.5.4 Acceptance tests.

4.5.4.1 Examination samples. Samples selected as specified in 4.5.1.2 shall be examined as specified in 4.5.2.

4.5.4.2 Test procedure. Each assembly selected in accordance with 4.5.1.3 shall be subjected to all acceptance tests specified in table II.

4.5.4.3 Failure. Any assembly that fails to pass any acceptance test, or exhibits defects in excess of AQL specified, shall be rejected. Rejected units may be reworked or repaired and resubmitted for acceptance testing.

4.5.5 Control tests. Assemblies for control tests shall be selected in accordance with 4.5.5.1 and subjected to tests specified in table II (see 6.5).

4.5.5.1 Frequency (1/500). One assembly shall be selected from each lot, consecutively produced except that not more than one shall be selected in a one (1) month period, nor less than one test in a three (3) month period.

4.5.5.1.1 Test failure. Failure of the assembly to meet the specified control tests shall be considered cause for rejection of production quantity represented. When cause for failure is determined, necessary corrections shall be made and proven by testing three randomly selected assemblies from that production quantity. One of three assemblies shall be subjected to all

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specified tests. Control test verification for the two remaining assemblies shall be limited to parameters directly related to failure cause and parameters affected by corrective action taken.

4.6 Methods of inspection.

4.6.1 Coil resistance. To determine conformance to 3.5.1, resistance of the solenoid valve coil, when measured between solenoid connector and ground, shall be not less than 60 ohms nor more than 70 ohms.

4.6.2 Pull-in voltage. To determine conformance to 3.5.2, supply oil conforming to MIL-L-21260, Grade 10, which conforms to MIL-L-2104, Grade 10, with 135 to 270 ppm biocide, at 250 ± 10^0F through a feed orifice 0.037 ± 0.001 inch diameter by 0.25 ± .02 inch long under pressure of 90 ± 5 psi, to verify that valve opens at not more than 18 volts and recloses when voltage is reduced to not less than 1 volt.

4.6.3 Bleed rate. To determine conformance to 3.5.3, after energizing valve to 30 ± 1 volt, supply oil conforming to MIL-L-21260, Grade 10, which conforms to MIL-L-2104, Grade 10, with 135 to 270 ppm biocide, at 250 ± 10^0F through a feed orifice 0.037 ± 0.001 inch diameter by 0.25 ± .02 inch long under pressure of 150 ± 5 psi, to verify that pressure in the cavity between orifice and valve does not exceed 18 psi.

TABLE II. Classification of tests.

Title	Requirements	Preproduction /Initial Prod. tests	Acceptance Tests – 100 percent	Control Tests - 1/500
Coil resistance	3.5.1	4.6.1	4.6.1	
Pull-in voltage	3.5.2	4.6.2	4.6.2	
Bleed rate	3.5.3	4.6.3	4.6.3	
Valve leakage	3.5.4	4.6.4	4.6.4	
Planar function	3.5.5	4.6.5		4.6.5
Endurance	3.5.6.1	4.6.6.1		4.6.6.1
Low temperature	3.5.7.1	4.6.7.1		4.6.7.1
High temperature	3.5.7.2	4.6.7.2		4.6.7.2
Thermal shock	3.5.7.3	4.6.7.3		4.6.7.3
Basic shock	3.5.7.4	4.6.7.4		4.6.7.4
Gun firing shock	3.5.7.5	4.6.7.5		4.6.7.5
Vibration, sinsuoidal	3.5.7.6	4.6.7.6.1		4.6.7.6.1
Vibration, random	3.5.7.7	4.6.7.6.2		4.6.7.6.2.1

4.6.4 Valve leakage. To determine conformance to 3.5.4, with solenoid in de-energized condition, supply oil conforming to MIL-L-21260, Grade 10, which conforms to MIL-L-2104, Grade 10, with 135 to 270 ppm biocide, at 250 ± 10^0F, through a 0.037 ± 0.001 diameter by 0.25 ±.02 inch long orifice under pressure of 150 ± 5 psi, to verify that valve leakage in a cavity between the orifice and valve does not exceed 10 psi.

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4.6.5 Planar function. To determine conformance to 3.5.5, valve, while submerged in oil conforming to MIL-L-21260, Grade 10, which conforms to MIL-L-2104, Grade 10, with 135 to 270 ppm and biocide, at 80 ± 5^0F, shall open at not more than 18 volts and reclose when voltage is reduced to not less than 2 volts in any of the planes specified in 3.5.5.

4.6.6 Physical characteristics.

4.6.6.1 Endurance. To determine conformance to 3.5.6.1, mount assembly with flange up or down or with bottom of flange vertical, mounting holes horizontal and the lead coming out the top. Supply oil conforming to MIL-L-21260, Grade 10, which conforms to MIL-L-2104, Grade 10, with 135 to 270 ppm biocide, at 250 ± 10^0F, through a 0.037 ± 0.001 inch diameter by 0.25 ± .02 inch long feed orifice under pressure of 150 ± 5 psi. Energize assembly with 30 ± 1 volts and subject assembly to endurance cycle specified in 3.5.6.1. After completion of endurance testing, assembly shall be tested in accordance with 4.6.1 through 4.6.5 (see 6.5).

4.6.7 Environmental conditions.

4.6.7.1 Low temperature. To determine conformance to 3.5.7.1, soak assembly at -60 ± 5^0F in oil conforming to MIL-L-46167 for 24 ± 1 hours. Stabilize oil bath temperature at -25 ± 5^0F. Supply assembly with oil conforming to MIL-L-46167, at -25 ± 5^0F through a 0.037 ± 0.001 inch diameter by 0.25 ± .02 inch long orifice under pressure of 90 ± 5 psi. Verify that the valve opens at not more than 18 volts and pressure in a cavity between orifice and valve does not exceed 30 psi. Also, verify that valve closes when voltage is reduced to not less than 1 volt and pressure drop across orifice does not exceed 10 psi. Assembly shall then be subjected to performance tests of 4.6.1 through 4.6.5, following which, verify that assembly lead wire does not crack when bent 180^0 ± round a 0.75 ± .02 inch diameter arbor (see 6.5).

4.6.7.2 High temperature. To determine conformance to 3.5.7.2, soak assembly at 300 ± 10^0F in oil conforming to MIL-L-21260, Grade 30, which conforms to MIL-L-2104, Grade 30, with 135 to 270 ppm biocide, for 24 ± 1 hours. Supply assembly with oil conforming to MIL-L-21260, Grade 30, which conforms to MIL-L-2104, Grade 30, with 135 to 270 ppm biocide, at 300 ± 10^0F through a 0.037 ± 0.001 inch diameter by 0.25 ± .02 inch long orifice under pressure of 90 ± 5 psi. Verify that valve opens at not more than 18 volts and pressure in a cavity between orifice and valve does not exceed 18 psi. Also, verify that valve closes when voltage is reduced to not less than 2 volts and pressure in a cavity between orifice and valve is not less than 80 psi. Assembly shall then be subjected to performance tests of 4.6.1 through 4.6.5 following which, verify that assembly lead wire does not crack when bent 180^0 ± 5^0 around a 0.75 ± .02 inch diameter arbor (see 6.5).

4.6.7.3 Thermal shock. To determine conformance to 3.5.7.3, subject assembly to 15 thermal shocks consisting of a non-energized soak for not less than 2 hours at -60 ± 5^0F and a soak of not less than 1 hour at 300 ± 5^0F while energized at +30 ± 1 volt. Measure the resistance

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of the solenoid valve coil prior to starting test and after each thermal cycle while at 70 $\pm 5^0$F. Test shall be performed in stagnant air with no oil flow to assembly. At conclusion of the test, assembly shall be tested in accordance with 4.6.1 through 4.6.5 (see 6.5).

4.6.7.4 Basic shock. To determine conformance to 3.5.7.4, assembly shall be mounted using the installed interface configuration for its intended application (i.e. hard mounted on a one to two inch steel plate) or equivalent and subject to test of MIL-STD-810, method 516.1, procedure 1. A total of three half sine wave shock pulses shall be applied in both directions along each of three mutually perpendicular axes. Peak amplitude shall be 30 $\pm$ 3g, 11 $\pm$ 1.1g milliseconds (ms). During exposure, monitor assembly and verify intermittent operation does not occur. At conclusion of test, examine assembly for damage and subject assembly to tests of 4.6.1 through 4.6.5 (see 6.5).

4.6.7.5 Gun firing shock. To determine conformance to 3.5.7.5, assembly shall be mounted using the installed interface configuration for its intended application (i.e. hard mounted on a one to two inch thick steel plate) or equivalent and subjected to test of MIL-STD-810, method 516.1, procedure IV. Subject assembly to gun firing shock of 55 $\pm$ 5.0 g for 2.5 $\pm$ .02 ms duration applied along the equivalent transmission transverse and longitudinal axes. During exposure, monitor assembly and verify intermittent operation does not occur. At conclusion of test, examine assembly for damage and subject assembly to tests of 4.6.1 through 4.6.5 (see 6.5).

4.6.7.6 Vibration. The assembly shall be mounted using the installed interface configuration for its intended application (i.e. hard mounted on a one to two inch thick steel plate or equivalent, capable of transmitting the vibration conditions specified).

4.6.7.6.1 Sinusoidal vibration. To determine conformance to 3.5.7.6, assembly shall be subjected to approximately 336 minutes of vibration, dwell and cycling in each of three mutually perpendicular axes. Cross axis vibration shall not exceed the input vibration while testing in any one of the three test axes. Assembly shall be scanned over the frequency range of 10 Hz to 2700 Hz at 1 to 2 g's input vibration, both de-energized and energized with +30 $\pm$ 1 volts, along each of the three mutually perpendicular axes. Utilizing micro-miniature accelerometers, vibratory response shall be monitored and plotted on a g versus Hz graph. As a minimum, response at or near the free end of assembly shall be recorded in the three axes of vibratory excitation. Where practical, resonance of any sub-component shall also be documented. Sinusoidal resonance search at cycle test level and rate may be used; however, search time shall then be included as a portion of total cycle time. Cycle testing shall be conducted as specified in 3.5.7.6. At the conclusion of this test, assembly shall be examined for damage, and subjected to tests of 4.6.1 through 4.6.5 (see 6.5).

4.6.7.6.2 Random vibration. To determine conformance to 3.5.7.7, nine assemblies shall be simultaneously subjected to 100 hours of random vibration durability testing as specified in 3.5.7.7 (i.e., three valves in each of three mutually perpendicular axes). Each assembly shall be electrically cycled as specified in 3.5.7.7 and monitored for continuity throughout the durability

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test. A non-removal, visual inspection shall be made at least every four hours during the test. At conclusion of test, each assembly shall be thoroughly examined for evidence of structural failure or excessive wear and subjected to tests of 4.6.1 through 4.6.5.

4.6.7.6.2.1 Control test. Perform the test of 4.6.7.6.2 except that the test duration time for the random vibration durability test shall be 25 hours (see 6.5).

4.7 Preparation for delivery. Assembly shall be inspected by supplier prior to shipment to ensure conformance to preservation, packaging, packing and marking requirements in section 5. Sampling inspections shall be limited to those characteristics that can be determined by visual examination.

5. PACKAGING

5.1 Preservation, packaging, packing, and marking. Preservation, packaging, packing, and marking for the desired level of protection shall be in accordance with the applicable packaging standard or packaging data sheet specified by the contracting authority (see 6.2).

6. NOTES

6.1 Intended use. The solenoid valve assembly is intended to provide proper oil pressures to the control valves used in the transmission of the Abrams tank.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Whether preproduction samples shall be furnished (see 3.1).
c. Whether initial production units shall not be furnished (see 3.2).
d. Whether responsibility for inspection equipment shall be other than as specified (see 4.1.1).
e. Whether test conditions shall be other than as specified (see 4.3.1).
f. Whether oil used in bench testing shall be other than as specified (see 4.3.1.3).
g. Selection of applicable level and packaging standard or packaging data sheet (see 5.1).

6.3 Preproduction. When a preproduction sample inspection is required, the item will be tested as specified in 4.4.1. The preproduction samples should consist of twenty units. The contracting officer should include specific instructions in all acquisition documents regarding arrangements for examination, tests and approval of the preproduction samples (see 3.1).

6.4 Process average. Sampling may be initiated if process average value for the first twenty assemblies inspected is less than AQL specified in 4.5.2.1 for major and minor defects.

$$\text{Process average} = \frac{\text{Number of defects}}{\text{Number of assemblies inspected}} \times 100$$

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If computed process average exceeds specified AQL, 100 percent inspection shall be performed and continued until such time that process average for twenty consecutive assemblies is less than the specified AQL.

6.5 Disposition of control test assemblies. Assemblies subjected to control tests in accordance with 4.6.6.1, 4.6.7.1, 4.6.7.2, 4.6.7.3, 4.6.7.4, 4.6.7.5, 4.6.7.6.1 and 4.6.7.6.2.1 shall not be used for any other purpose and shall be indelibly, marked "DO NOT USE".

6.6 Recycled materials. Use of recycled materials which meet the requirements of applicable material specifications without jeopardizing intended use of the item shall be encouraged (see 3.3)

6.7 Supersession data. This specification supersedes Allison Division, General Motors Corporation, internal specification DAX13394F, dated 15 December 1982.

Custodian:
Army - AT

Preparing activity:
Army - AT

Project No. 2520-A206

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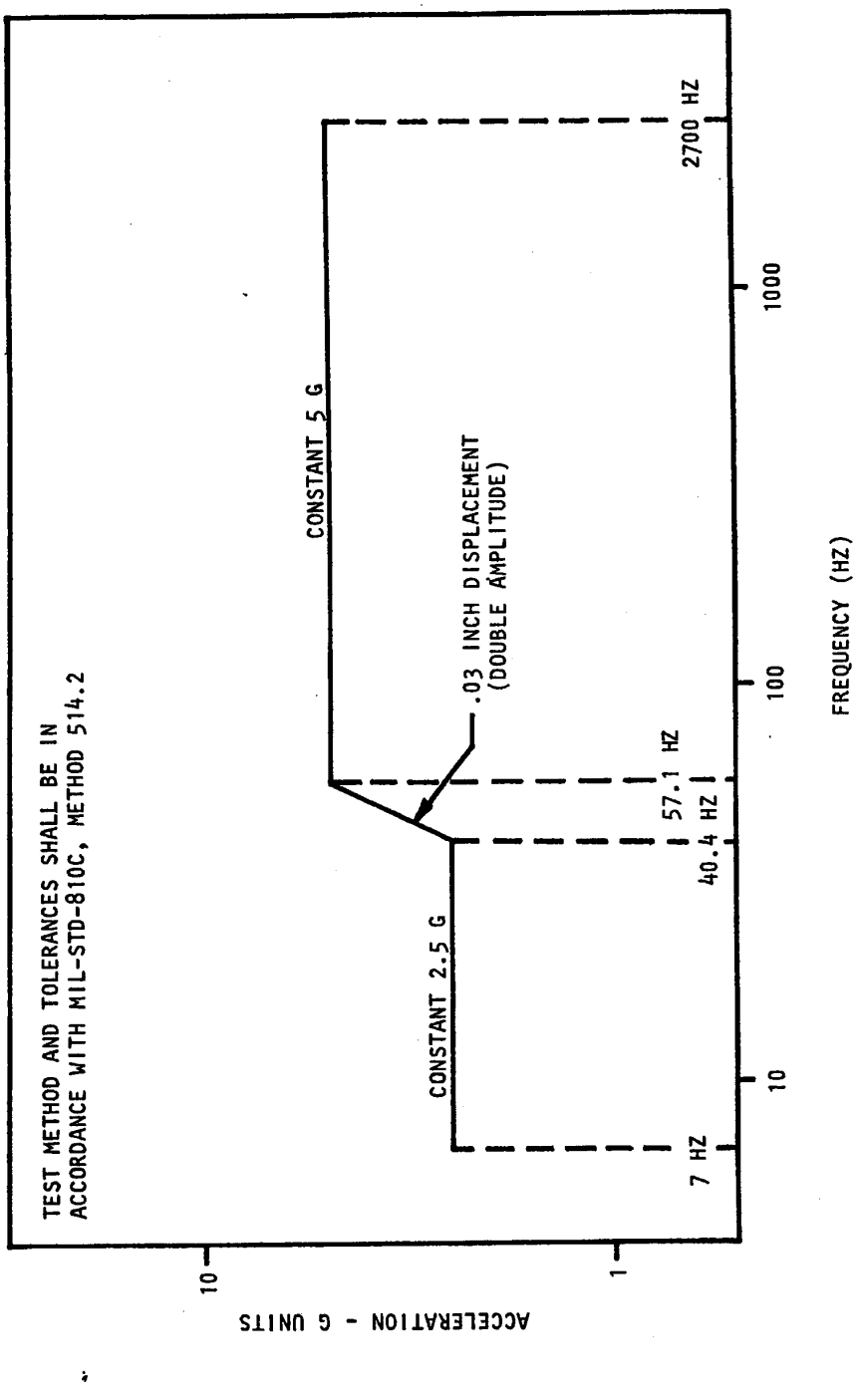


FIGURE 1. Solenoid valve assembly sinusoidal vibration test schedule.

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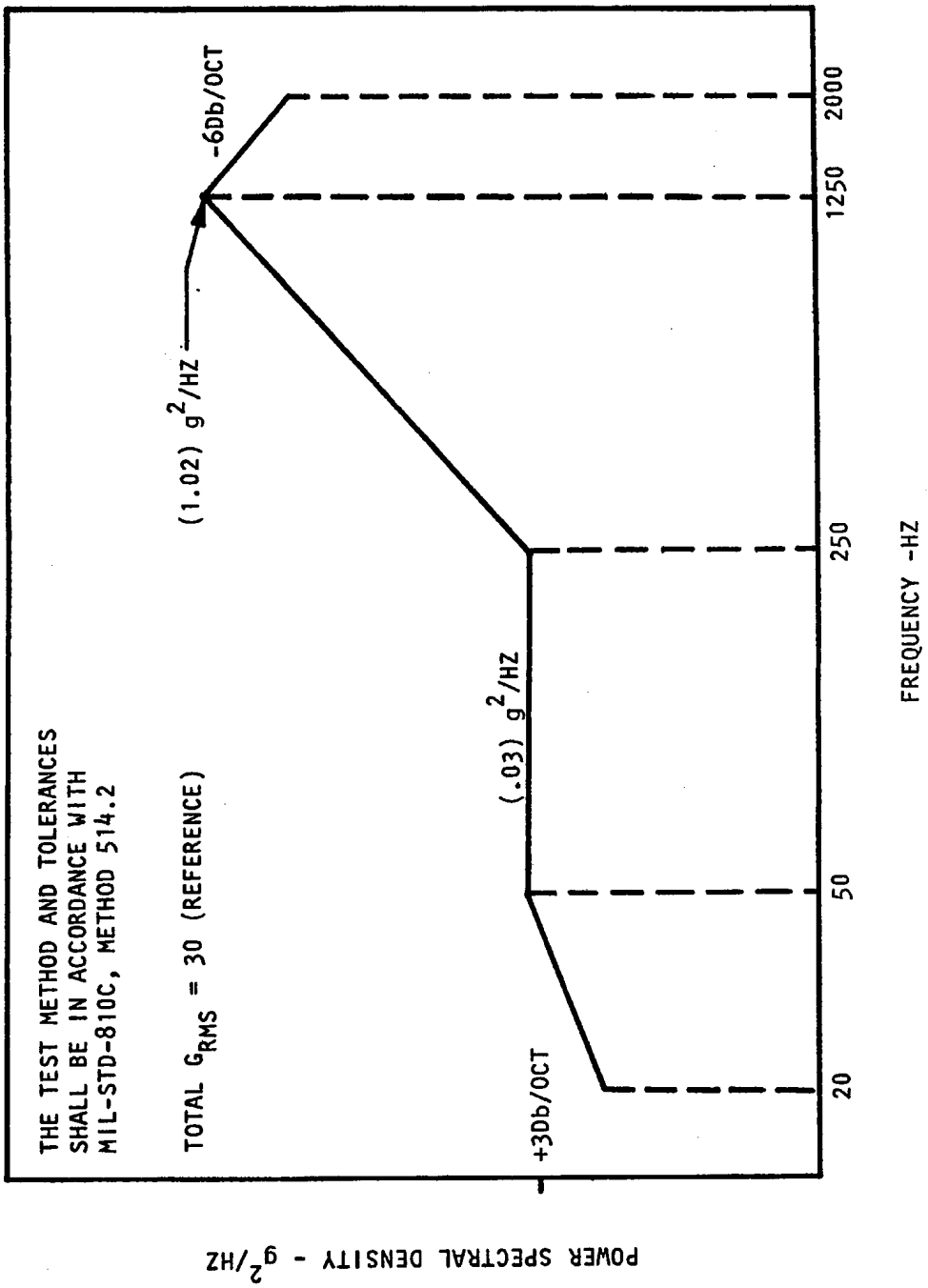


FIGURE 2. Solenoid valve assembly random vibration test spectrum.

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